

1. Define Computer Network

A computer network is a set of interconnected computers that share resources and data. The connections between the computers can be established using wired or wireless media. Computer networks enable communication and resource sharing among devices, improving efficiency and collaboration.

2. What is DNS, and how does it work?

The Domain Name System (DNS) translates domain names (like `www.example.com`) into IP addresses (like `192.0.2.1`) that computers use to identify each other on the network. When a user enters a domain name into a browser, the DNS server looks up the corresponding IP address and directs the browser to the correct site. DNS consists of a hierarchical structure of servers, including root servers, top-level domain servers, and authoritative name servers.

3. What is a MAC address, and how is it different from an IP address?

A Media Access Control (MAC) address is a unique identifier assigned to a network interface card (NIC) for communications on the physical network segment. It is a hardware address that is used in the data link layer of the OSI model. An IP address, on the other hand, is a logical address assigned to a device in a network for identifying and locating the device within a network. The IP address is used in the network layer of the OSI model.

4. Compare the following networking devices - Hub and Switch.

Hub: A hub is a basic networking device that connects multiple computers in a network. It operates at the physical layer and broadcasts data to all connected devices, regardless of the destination.

Switch: A switch operates at the data link layer and connects multiple devices in a network. It uses MAC addresses to forward data only to the specific device for which the data is intended, reducing unnecessary traffic and improving network efficiency.

5. What is Error detection in Computer network?

Error detection is the process of identifying errors in transmitted data over a network. This can be achieved through various techniques such as parity checks, checksums, and cyclic redundancy checks (CRC). These methods help ensure the integrity of the data by detecting and possibly correcting errors that occur during transmission.

6. Compare the following networking devices - Router and Bridge.

Router: A router operates at the network layer and is used to connect multiple networks together. It routes data from one network to another and determines the best path for data transmission.

Bridge: A bridge operates at the data link layer and connects two or more network segments, making them function as a single network. It filters traffic between segments based on MAC addresses to reduce collisions and improve network performance.

7. What is Wireless Transmission Medium?

A wireless transmission medium refers to the use of electromagnetic waves to transmit data through the air without the need for physical cables. Common wireless transmission media include radio waves, microwaves, and infrared waves. Wireless networks use this medium to connect devices such as laptops, smartphones, and tablets to the network.

8. Describe two common methods used for error detection in the Data Link Layer and briefly explain how they work.

Parity Check: This method involves adding a parity bit to the data to make the number of 1's either even (even parity) or odd (odd parity). The receiver checks the parity bit to detect errors in

the transmitted data.

Cyclic Redundancy Check (CRC): CRC is a more complex method that uses polynomial division to detect errors. The data is divided by a predetermined polynomial, and the remainder is appended to the data as a CRC code. The receiver performs the same division and checks if the remainder matches the CRC code.

9. What is shortest path in network layer?

The shortest path in the network layer refers to the path between two nodes (e.g., routers) in a network that has the minimum total distance or cost. Shortest path algorithms, such as Dijkstra's algorithm and the Bellman-Ford algorithm, are used to determine the most efficient route for data packets to travel across the network.

10. Explain the concept of packet switching in data communication.

Packet switching is a method of data transmission in which data is broken into small packets before being sent over the network. Each packet is transmitted independently and can take different paths to reach the destination. Once all packets arrive, they are reassembled into the original message. This method improves network efficiency and reduces latency.

11. Discuss the OSI model in detail, including the functions of each layer and examples of protocols used in each layer. Explain CSMA in medium access layer.

The OSI (Open Systems Interconnection) model is a conceptual framework used to understand network interactions in seven layers:

Physical Layer: Deals with the physical connection between devices and the transmission and reception of raw bit streams. Examples: Ethernet, USB.

Data Link Layer: Provides error detection and correction, and controls how data is placed on the network. Examples: MAC addresses, Ethernet.

Network Layer: Manages data routing, forwarding, and addressing. Examples: IP, ICMP.

Transport Layer: Ensures reliable data transmission with error recovery and flow control. Examples: TCP, UDP.

Session Layer: Manages sessions and controls dialogs between applications. Examples: NetBIOS, PPTP.

Presentation Layer: Translates, encrypts, and compresses data. Examples: JPEG, SSL/TLS.

Application Layer: Provides network services to applications. Examples: HTTP, FTP, SMTP.

CSMA (Carrier Sense Multiple Access) is a protocol used in the medium access layer to control access to the network. It ensures that devices check the network for other traffic before transmitting data to avoid collisions.

12. Explain the concept of network security, including encryption techniques, authentication methods, and intrusion detection systems.

Network security encompasses measures taken to protect data during transmission and to prevent unauthorized access. Key components include:

Encryption Techniques: Methods like AES, RSA, and SSL/TLS are used to encrypt data, making it unreadable to unauthorized users.

Authentication Methods: Techniques like passwords, biometrics, and multi-factor authentication (MFA) are used to verify user identities.

Intrusion Detection Systems (IDS): Tools that monitor network traffic for suspicious activities and alert administrators to potential threats.

13. Describe the differences between wired and wireless networks, including their technologies, advantages, and limitations.

Wired Networks: Use physical cables (e.g., Ethernet) for data transmission. Advantages include higher speed, reliability, and security. Limitations include limited mobility and higher installation costs.

Wireless Networks: Use electromagnetic waves for data transmission. Advantages include mobility, ease of installation, and flexibility. Limitations include potential interference, lower speeds, and security vulnerabilities.

14. Explain Switch network device.

A switch is a networking device that connects devices within a network and uses MAC addresses to forward data only to the intended recipient. This reduces collisions and improves network performance compared to hubs. Switches operate at the data link layer of the OSI model.

15. Explain data link layer in OSI layer.

The data link layer is the second layer in the OSI model. It provides error detection and correction, frames data for transmission, and controls access to the physical medium. It includes two sublayers: the Logical Link Control (LLC) and the Media Access Control (MAC).

16. What is TCP/IP protocol?

The TCP/IP protocol suite is a set of communication protocols used to interconnect network devices on the internet. It includes:

TCP (Transmission Control Protocol): Ensures reliable, ordered, and error-checked delivery of data.

IP (Internet Protocol): Handles addressing and routing of data packets.

17. Explain Router network device.

A router is a device that connects multiple networks and forwards data packets between them based on their IP addresses. Routers determine the best path for data transmission and are essential for directing internet traffic.

18. Explain Physical Layer. What are Primary functions of physical layer?

The physical layer is the first layer of the OSI model. It deals with the transmission of raw bit streams over a physical medium. Primary functions include:

Defining hardware specifications

Encoding and signaling

Data transmission and reception

Topology and network design

19. Explain Application Layer. What the function of application layer?

The application layer is the seventh layer of the OSI model. It provides network services directly to end-user applications. Functions include:

Identifying communication partners

Ensuring data integrity

User authentication and privacy

Examples of application layer protocols: HTTP, FTP, SMTP.

20. Explain the process of IP addressing, including IPv4 and IPv6, subnetting, and IP classes.

IPv4: Uses 32-bit addresses, divided into four octets. Classes range from A to E based on address range.

IPv6: Uses 128-bit addresses, allowing for a vastly larger number of unique addresses.

Subnetting: Dividing a network into smaller sub-networks to improve management and security.

IP Classes:

Class A: 0.0.0.0 to 127.255.255.255

Class B: 128.0.0.0 to 191.255.255.255

Class C: 192.0.0.0 to 223.255.255.255

Class D: 224.0.0.0 to 239.255.255.255 (Multicast)

Class E: 240.0.0.0 to 255.255.255.255 (Reserved)

21. Write down the full forms of:

SSH: Secure Shell

NTP: Network Time Protocol

OSI: Open Systems Interconnection

TCP: Transmission Control Protocol

ICMP: Internet Control Message Protocol

ARP: Address Resolution Protocol

LLC: Logical Link Control

MAC: Media Access Control

22. Define the Following: BRIDGE, HUB, SWITCH, ROUTER, and GATEWAY.

Bridge: Connects two or more network segments, reducing collisions and improving traffic management.

Hub: Connects multiple devices in a network, broadcasting data to all connected devices.

Switch: Connects devices within a network, using MAC addresses to forward data only to intended recipients.

Router: Connects multiple networks, directing data packets based on IP addresses.

Gateway: Acts as a bridge between different networks, often used to connect a local network to the internet.